

PROGRESO, Mexico

WMO: 765930

Lat: 21.2767N

Lon: 89.6539W

Elev: 7

StdP: 14.69

Time Zone: -6.00 (MEX)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 62.8	(c) 65.4	(d) 53.8	(e) 61.6	(f) 68.9	(g) 56.2	(h) 67.3	(i) 69.9	(j) 25.5	(k) 72.8	(l) 20.7	(m) 72.7	(n) 4.5	(o) 140	(p) 0.347

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.2	(c) 94.3	(d) 82.0	(e) 92.0	(f) 81.2	(g) 90.0	(h) 81.1	(i) 85.8	(j) 90.8	(k) 84.6	(l) 89.1	(m) 83.6	(n) 88.0	(o) 10.8	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
84.5	181.1	90.1	83.3	174.3	88.6	82.3	168.5	87.5	50.3	90.8	48.7	89.2	47.6	88.0	93.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.4	(b) 18.6	(c) 16.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 21.4	(b) 18.6	(c) 16.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	79.4	73.7	75.6	77.8	80.3	81.9	82.4	82.0	82.1	81.9	80.8	77.8	75.7
	DBStd	4.14	3.47	3.41	4.18	3.85	2.73	2.29	1.60	1.81	1.78	2.25	3.05	2.98
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10720	734	717	862	910	990	973	993	997	957	956	834	797
	CDD65	5245	269	297	397	460	525	523	528	532	507	491	384	332
	CDH74	50317	1143	1783	3225	4551	5797	5879	6034	6026	5716	5167	3111	1885
	CDH80	15307	118	354	937	1466	2004	2123	2077	2100	1855	1452	544	277
Wind	WSAvg	7.7	6.7	7.8	9.7	8.1	8.6	7.1	7.8	7.3	6.2	6.9	8.5	7.5
Precipitation	PrecAvg	16.2	0.7	0.7	0.3	0.4	0.7	3.2	1.8	2.1	2.9	2.2	1.1	0.8
	PrecMax	29.2	4.6	3.7	2.8	3.1	3.3	11.5	4.6	7.1	8.5	9.1	4.8	4.9
	PrecMin	4.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.0	0.0	0.0	0.0
	PrecStd	6.8	1.1	1.0	0.6	0.7	0.8	3.0	1.3	1.9	2.4	2.4	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.7	91.6	94.9	97.9	96.4	95.5	93.6	94.1	92.9	92.2	88.8	89.0
		MCWB	74.6	75.6	82.1	83.6	81.6	83.3	83.5	83.0	85.6	82.8	81.4	76.1
	2%	DB	83.3	87.9	91.1	93.2	92.4	91.9	90.4	90.5	89.0	89.1	86.4	85.7
		MCWB	74.0	75.9	79.3	79.8	81.2	81.2	83.1	82.6	83.0	81.2	79.5	77.6
	5%	DB	80.7	84.8	88.0	89.8	89.5	89.2	88.9	88.7	87.8	87.5	84.6	82.5
		MCWB	73.5	75.6	78.2	78.6	80.7	81.4	82.6	82.2	82.4	80.8	78.3	76.0
	10%	DB	79.0	82.2	84.7	86.8	87.5	87.5	87.5	87.5	86.4	85.9	83.1	80.6
		MCWB	73.2	75.1	77.3	77.9	80.1	81.1	81.8	81.8	81.1	79.9	77.5	75.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.3	81.5	85.3	87.8	87.1	86.6	87.0	85.8	86.1	84.8	84.1	80.2
		MCDB	82.6	87.7	92.7	95.0	94.0	91.9	91.6	90.4	90.8	89.4	87.9	84.9
	2%	WB	75.5	78.9	82.7	82.8	84.0	84.5	85.0	84.3	84.4	83.2	81.5	78.2
		MCDB	80.3	83.3	88.8	88.9	88.9	88.7	89.0	88.7	88.6	87.5	84.8	82.6
	5%	WB	74.7	77.1	80.4	80.5	82.3	82.9	83.7	83.3	83.1	81.7	79.7	77.1
		MCDB	79.3	81.5	84.9	86.9	87.4	87.1	87.7	87.7	87.1	86.0	83.0	81.0
	10%	WB	73.9	75.6	78.6	78.9	80.8	81.6	82.4	82.1	81.8	80.4	78.3	76.0
		MCDB	78.3	80.2	83.0	84.9	86.3	86.0	86.8	86.7	85.6	84.8	81.8	79.9
Mean Daily Temperature Range	5% DB	MDBR	9.8	11.1	13.9	12.9	11.6	10.9	10.3	10.2	9.0	9.6	9.9	10.2
		MCDBR	13.5	17.9	19.7	19.3	16.5	14.9	14.8	14.2	12.6	13.0	12.4	14.3
		MCWBR	7.3	8.9	10.1	9.4	8.4	7.7	8.8	7.6	7.9	7.1	7.0	7.5
	5% WB	MCDBR	10.7	15.7	18.0	18.0	15.1	13.7	14.1	13.2	13.3	12.6	10.7	11.8
		MCWBR	6.9	9.5	11.8	10.6	9.4	8.9	9.6	8.6	8.9	8.2	7.5	7.6
Clear-Sky Solar Irradiance	taub	0.362	0.376	0.401	0.436	0.451	0.449	0.465	0.442	0.411	0.397	0.376	0.373	
	taud	2.565	2.520	2.417	2.299	2.310	2.373	2.308	2.391	2.462	2.506	2.525	2.537	
	Ebn at Noon	290	291	287	277	270	268	264	271	280	281	282	281	
	Edh at Noon	30	33	37	43	42	39	42	38	36	33	30	30	
All-Sky Solar Radiation	RadAvg	1301	1571	1860	2057	2096	1936	2098	2046	1841	1617	1404	1265	
	RadStd	96	122	87	98	100	111	76	64	95	73	68	80	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	+1.58	+1.91	N/A	N/A	N/A	N/A

Nomenclature: See separate page